SIEMENS

Industrial Edge

Modbus TCP Connector

Manual

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Preface

1

1.1 Introduction

1.1.1 Introduction to Modbus TCP Connector

Welcome to the Modbus TCP Connector documentation for Industrial Edge.

The Modbus TCP Connector connects various devices via the Modbus TCP protocol to the Industrial Edge system.

Connectors collect data from PLCs and devices and make it available on the Industrial Edge system. The Modbus TCP Connector accesses data via the Modbus TCP protocol.

This connector is part of the Connectivity Suite (CS) and supports standardized configuration and data transfer.

The Common Configurator handles the connector configuration. It is available free of charge in the Industrial Edge marketplace.

Industrial Edge App developers and users can integrate the data into their own applications and services. The Industrial Edge Connectors provide a unified data interface that reduces development effort and forms the basis for modularized software solutions.

The Modbus TCP Connector supports reading and writing values for devices via the Modbus TCP protocol. The following devices are supported:

- Schneider Quantum TCP/IP 140 CPU 113 03
- Schneider TSX Premium TSXP573634
- Schneider TSX Momentum 171CCC98020 (TCP/IP)
- Schneider Micro TSX 3710001
- Schneider M 340
- Schneider TM218LDAE24DRHN
- Schneider TM238LDD24DT
- Schneider TM258LD42DT
- Siemens SIWAREX WP231

- Siemens SIWAREX WP241

The Modbus TCP Connector includes the following key features:

Feature	Description
Compatibility	Interfaces with any Modbus TCP PLC using the standard protocol.
Direct Integration	As part of the Connectivity Suite, enables direct availability of Modbus TCP tag values in IIH without needing the Databus.
Real-time data subscription and write-back	Enables live data exchange with Modbus TCP-compatible devices.
Comprehensive data type support	Supports all data types, excluding arrays and structures.
Driver status feedback	Provides driver status messages directly in the Common Configurator.

1.1.2 Validity of the documentation

This documentation covers the installation and configuration of Modbus TCP Connector on Industrial Edge Devices (IE Devices). It provides step-by-step guidance for setting up and using the app.

1.1.3 Overview of additional documentation

The following resources provide additional information to complement this manual.

Documentation	Main contents
Industrial Edge Marketplace	Purchase app licenses
Industrial Edge Hub	Describes the functions of the Industrial Edge platform and the Edge Management System. All documentation for the Industrial Edge platform is available under Documents.
System overview	Provides an overview of all Industrial Edge solutions.
Industrial Edge trainings	Web-based training to integrate IT in production with Industrial Edge
Databus	Documentation for Databus
Industrial Information Hub (IIH)	Manual for Industrial Information Hub
SIMATIC SCADA Export for TIA Portal	Information and download for the SIMATIC SCADA export

1.2 Legal information

1.2.1 Cybersecurity information

Siemens provides products and solutions with industrial cybersecurity functions that support the secure operation of plants, systems, machines and networks.

In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art industrial cybersecurity concept. Siemens' products and solutions constitute one element of such a concept.

Customers are responsible for preventing unauthorized access to their plants, systems, machines and networks. Such systems, machines and components should only be connected to an enterprise network or the internet if and to the extent such a connection is necessary and only when appropriate security measures (e.g. firewalls and/or network segmentation) are in place.

For additional information on industrial cybersecurity measures that may be implemented, please visit https://www.siemens.com/cybersecurity-industry.

Siemens' products and solutions undergo continuous development to make them more secure. Siemens strongly recommends that product updates are applied as soon as they are available and that the latest product versions are used. Use of product versions that are no longer supported, and failure to apply the latest updates may increase customer's exposure to cyber threats.

To stay informed about product updates, subscribe to the Siemens Industrial Cybersecurity RSS Feed under https://new.siemens.com/cert.

1.2.2 Security Information for Industrial Edge Apps

Security information (assumptions/constraints) for Industrial Edge Apps is as follows:

- Only authorized internal operators will have access to IE Device within a secure network using VPN connection.
- Perimeter firewall configuration responsibility lies with the end customer.
- The security guidelines for usage of USB Flash Drives in the shop floor area are applied accordingly.
- Creating users with appropriate access rights upon commissioning is the responsibility of the operator.
- The customer is responsible for configuring the application on the basis of the system requirements and technical capabilities of the documented App according to the Installation / User Manual such that the automation system performance is not impacted.
- The system is installed in an environment ensuring that physical access is limited to authorized maintenance personnel only. Managing unauthorized attachment of removable devices is the responsibility of the operator.

- The platform including hardware, firmware and operating system is securely configured and maintained by the operator.
- The operator is capable of protecting the environment from malware infection.
- Centralized IT security components (Active Directory, Centralized IT Logging Server) are provided and well secured by the operator and are trustworthy.
- The operator personnel accessing the system is well trained in the usage of the system and general information security aspects like password handling, removable media, etc.
- The operator is responsible for the CIA (Confidentiality, Integrity and Availability) of data stored outside the IE Device.
- The operator is responsible for configuring the CPUs with appropriate read/write access levels (legitimization), and for configuring the Industrial Edge Apps using appropriate passwords for data collection from CPUs.
- The customer takes care about the time synchronization of Industrial Edge Management (IEM) and IE Device.

1.2.3 General Data Protection Regulation (GDPR)

Siemens observes the principles of data protection, in particular the principle of data minimization (privacy by design). For the Modbus TCP Connector product, this means that:

1.2.3.1 Personal data

There is no personal data collected but following data is stored to allow machine to machine communication:

- Databus access data
- Tag data and metadata of field devices

1.2.3.1.1 Protection of personal data

If you link the above data with other data (e.g. shift schedules) or if you store personal data on the same medium (e.g. hard disk) and thus create a personal reference, you must ensure that the data protection guidelines are observed.

1.2.3.2 Purpose

The above data is required for the following purposes:

- Access protection and security measures
- Reporting system for traceability and availability

The storage of data is appropriate and limited to what is necessary, as it is essential to identify the authorized operators.

1.2.3.3 Securing data

The above data will not be stored anonymously or pseudonymized, because the purpose (identification of the operating personnel) cannot be achieved otherwise.

The following data will only be used within the product and Edge ecosystem and will not be automatically shared with third parties or unauthorized individuals:

- Databus access data
- Modbus TCP authentication data

The above data is protected against loss of integrity and confidentiality by state-of-the-art security measures.

Tag data and metadata of field devices is used only within the product and is not automatically shared with third parties or unauthorized individuals.

You must ensure access protection as part of your process configuration.

What's New

2

2.1 What's new in Modbus TCP Connector

2.1.1 Version 4.2.0

2.1.1.1 Features

- Support hot configuration deployment, allowing you to deploy configuration changes without restarting the connector, minimizing downtime and ensuring continuous operation
- Support auditable events for security monitoring and compliance
- Reduce logging for the ARM platform to error level by default. You can change the log level if needed. See Logging.

2.1.1.2 Bug fixes

- Fix reading issues for some devices
- Update the version of communication library

2.1.2 Version 4.1.0

2.1.2.1 Features

- Support for Read File Record function (0x14), for example, historical records from SIEMENS PAC devices
- Support for ARM64 devices
- Support for Industrial Edge Memory Management, which allows reservation of memory and a larger limit to load very large configuration files. Requires AMD64 IE Device-OS version 3.0 or later, or IE Device version 1.21 or later.
- Support Raw data type, providing the capability to read continuous register memory areas

2.1.3 Version 4.0.0

2.1.3.1 Features

- Support for remote configuration from the IEM. Requires Common Configurator on IEM and IE Device-OS 3.0.0 or later.
- Support for cyclic continuous. Publishes tags in each cycle, even if they do not change.
- Improved log files. If a log repeats with only a timestamp change, the system does not log it instantly. Instead, it counts the occurrences and logs the entry after ten minutes.

2.1.3.2 Limitation

- Requires IE Device-OS 2.1 or later

Overview

3

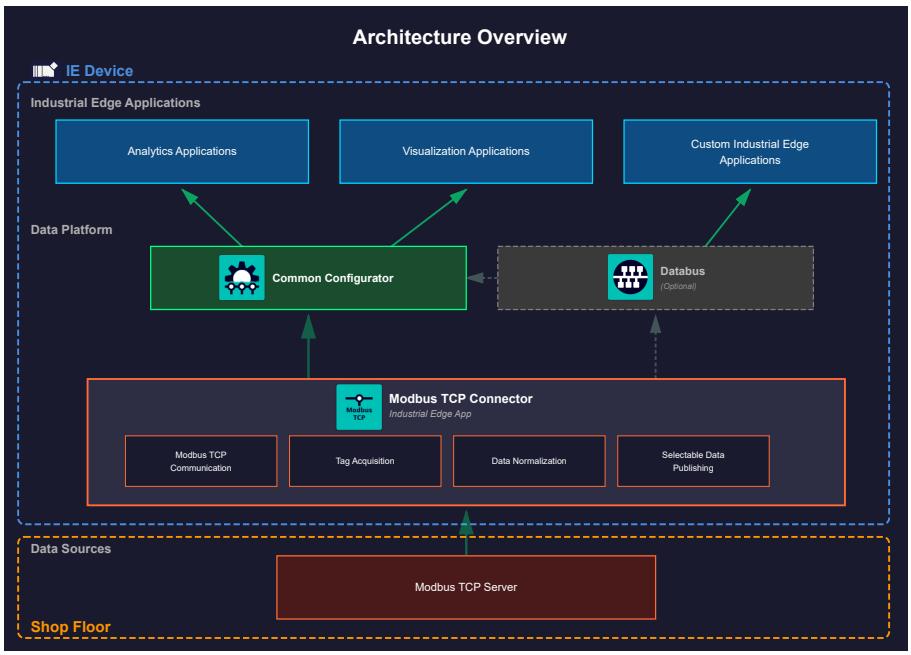
3.1 Overview of Modbus TCP Connector

Modbus TCP Connector is an app that runs on IE Devices. Install Modbus TCP Connector on IE Devices to connect to your Modbus TCP PLCs.

Modbus TCP Connector transfers tag values from Modbus TCP PLCs to the Databus or Industrial Information Hub (IIH). This enables you to analyze and visualize Modbus TCP tag values in apps such as Performance Insight.

Modbus TCP Connector supports any Modbus TCP PLC that uses the standard protocol.

As a Connectivity Suite (CS) Connector, Modbus TCP Connector also makes tag values available directly in the IIH without using the Databus. For more information, see IIH documentation.



3.2 Supported Data Types

3.2.1 Supported data types in Modbus TCP Connector

Data type	Description
Bool	1 bit
Int	Signed integer, 2 bytes
UInt	Unsigned integer, 2 bytes
DInt	Signed integer, 4 bytes
UDInt	Unsigned integer, 4 bytes
LInt	Signed integer, 8 bytes
ULInt	Unsigned integer, 8 bytes
Real	Floating point, 4 bytes
LReal	Floating point, 8 bytes
Word	Unsigned integer, 2 bytes
String	Character string. The maximum length is 250 characters.
Raw	A data block. Data length specification is achieved through dot notation in the address field (for example, address.length).

3.2.2 Default data types in Industrial Edge

The current version of the Common Configurator publishes Connectivity Suite data types as standard Industrial Edge data types. Alternative type configurations are not available.

The following data types map between the Connectivity Suite and Industrial Edge:

Connectivity Suite data type	Industrial Edge data type
Bool	Bool
UInt16	UInt
Int16	Int
UInt32	UDInt
Int32	DInt
UInt64	ULInt
Int64	LInt
Float32	Real
Float64	LReal
String	String
UInt16	Word

To convert Connectivity Suite data to the appropriate Industrial Edge data types, you can also configure the Databus manually. The configuration file is located in the "Management" tab of the IE Device.

Installation 4

4.1 System requirements

Listed below are the system requirements to ensure successful installation of the Modbus TCP Connector app.

4.1.1 Software requirements

You need Google Chrome Version 72 or later.

The apps are available in the Industrial Edge Marketplace. Transfer the latest app version to your Industrial Edge Management (IEM) system and install it on your IE Device.

App/Component	Required version	Description	Source
IE Device-OS	2.1 or later	Operating system for IE Devices	IEM System
Common Configurator	V2.0 or later	Configures data sources and data points on IE Devices	Industrial Edge Marketplace
IIH Semantics	V2.0 or later	Optional	Industrial Edge Marketplace
IIH Registry Service	V2.0 or later	Creates connections between components; required to view connectors in the overview	Industrial Edge Marketplace
Common Import Converter	V3.0 or later	Converts different file formats to IIH-compatible format; imports tag values from TIA Portal projects offline	Industrial Edge Marketplace
Modbus TCP Connector	V4.2	Mandatory	Industrial Edge Marketplace
Databus	V3.0 or later	Required if you want to publish tag values to the Databus	Industrial Edge Marketplace
SIMATIC SCADA Export for TIA Portal	TIA Portal V18 or later	Required if you want to import tag values from TIA Portal projects offline	Siemens Industry Online Support (SIOS)

> **NOTICE**
>
> All apps listed above must be installed on the same IE Device. The latest version of Databus V3.0.0 supports communication with connectivity suite connectors because the Databus Gateway application is now integrated into Databus.

4.1.2 Hardware requirements

You need the following:

- A device on which IEM is running
- An Industrial IE Device compatible with IEM:
  - Model: for example, SIMATIC IPC127E, IPC227E Nanobox, SIMATIC IPC427E, or IPC827E
  - Hard disk: at least 10 GB available
  - RAM: 2 GB available
- The Industrial IE Device must be onboarded to IEM

> **NOTICE**
>
> IEM, IE Devices, and web browsers must use synchronous UTC time zone settings.

4.1.3 Updating the Databus app

Update the Databus app to version 3.0.0 by following these steps:

1. Navigate to "Common Configurator" and create a backup of your current configuration.
2. Upgrade the "Databus" app to version 3.0.0.
3. Wait 3 minutes before continuing to ensure all processes are stable.
4. Uninstall the "Databus Gateway" app from your IE Device.

4.2 Installation process

4.2.1 Overview of the installation process

Industrial Edge apps are available in the Industrial Edge Hub Library. The Industrial Edge Hub (IE Hub) is the central repository for all available Industrial Edge apps (IE apps) from Siemens and other app partners in the ecosystem.

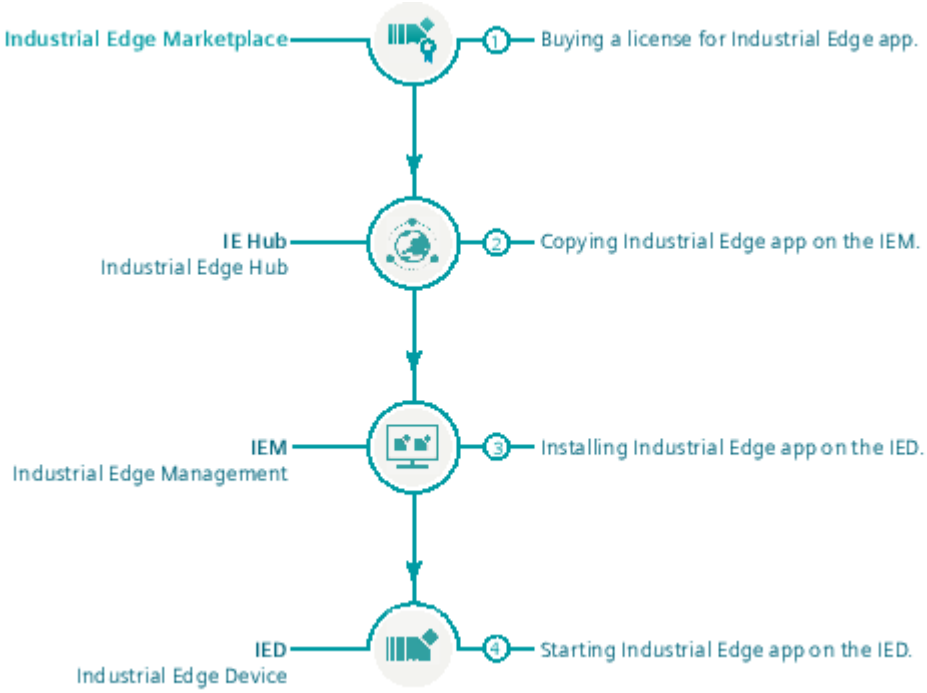
The installation of an IE app involves the following steps:

- Copying an IE app from the IE Hub to IEM
- Installing an app on one or more IE Devices
- Starting an installed app on an IE Device

The installation steps are demonstrated in the following video:

4.2.1.1 Overview of the entire IE app purchase process

The following figure shows the steps required to install and start an Industrial Edge app purchased from the Industrial Edge Marketplace on an IE Device:



4.2.1.2 Purchasing a license for an IE app

Purchase licenses for individual IE apps in the Industrial Edge Marketplace. For detailed instructions, see How to buy on the Siemens Industrial Edge Marketplace.

For more information on purchasing IE apps and managing your account, see Frequently Asked Questions.

4.2.2 Copying Modbus TCP Connector from the IE Hub to the IEM catalog

This section explains how to transfer an Industrial Edge app from the IE Hub to the IEM catalog of one or more IEM instances.

IEM is a control level for managing devices, apps, and users on a shop floor.

Before you start, ensure you have the following:

- Internet connection
- Access to the IE Hub
- At least one available IEM instance

4.2.2.1 Copying an app to the IEM catalog

Follow these steps:

1. Open the IE Hub and enter your credentials.
   The IE Hub home screen appears.
2. Open the "Library" tab in the Industrial Edge Hub.
   The library contains all apps for which you have purchased a license and all system apps.
3. Click the desired app.
   The app details appear.
4. In the drop-down list, select all IEM instances to which you want to copy the app.
5. Click "Copy latest version to IEM".
   The app is copied and a corresponding job is created. You can track the job status in the status window of the corresponding IEM instance.

4.2.3 Installing Modbus TCP Connector on the IE Device

This section explains how to install an Industrial Edge app from an IEM instance on an IE Device.

Before you start, ensure you have access to the IEM instance where the app is available in the catalog.

4.2.3.1 Installing the app on the IE Device

Follow these steps:

1. Open the IEM home page and log in.
2. Open the IEM catalog and click the app you want to install.
3. Click the "Install" button.
   The "Install App" dialog window appears. If you have a configuration file from another IE Device, you can upload it here.
4. Click "Next".
   The "Install App" dialog window appears. If you have a configuration file from another IE Device, you can upload it here.
5. Select one or more IE Devices where you want to install the app. The devices must be powered on and online.
6. Select one of the following installation options:
   - Click "Install Delayed" or "Install Later" to start the installation at a later time.
   - Click "Install Now" to install the app immediately.
7. Click the "Job Status" tab to view the installation progress.

After the installation completes, check "My Installed Apps" to see which apps are installed on which IE Devices.

> **Installation of other apps**
>
> All apps can be installed in the same way as explained above.

4.2.4 Starting Modbus TCP Connector on the IE Device

Before you start the Modbus TCP Connector app, ensure you have access to the IE Device on which the app is installed.

4.2.4.1 Starting the Modbus TCP Connector app

1. Open the IE Device home page and log in.
2. Open the "Apps" tab.
3. Launch the Common Configurator app and open the "Get Data" tab.
   The "Modbus TCP Connector" app appears in the available connectors list.
4. Click "Modbus TCP Connector" to start configuring the data sources and tags.

Migration

5

5.1 Upgrading Modbus TCP Connector

Modbus TCP Connector v2.1.0 uses Databus (MQTT), while the new Modbus TCP Connector (v3.0.0 or later) uses Connectivity Suite (gRPC). The new version offers significantly better performance. This process enables you to smoothly upgrade from v2.1.0 to v3.0.0 or later.

5.1.1 Prerequisites

Ensure the following components are installed before upgrading:

Component	Version	Function
Common Configurator	v1.11 or later	Configure data sources and tags
Common Import Converter	v2.1.0 or later	Import tags or convert configuration from v2.1.0 to v3.0.0
Registry Service	v1.8.0 or later	Service management

NOTICE

Ensure the Modbus TCP Connector app version is 2.1.0.

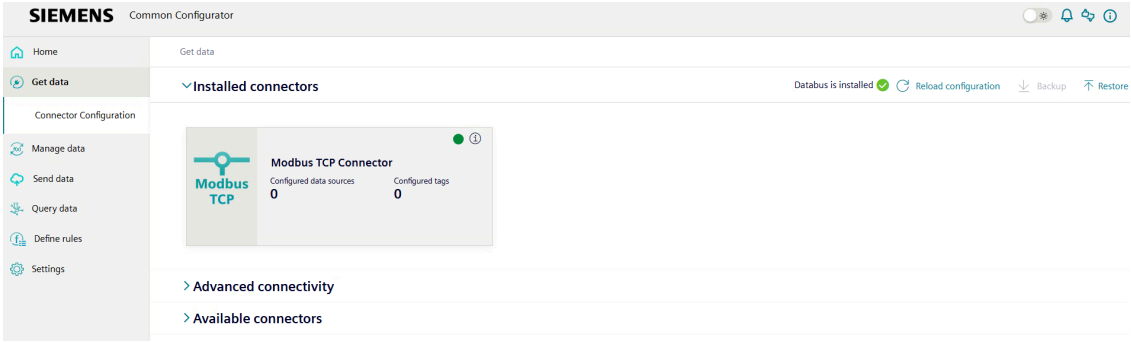
5.1.2 Procedure

Follow these steps to upgrade Modbus TCP Connector.

1. Copy the latest version from the Industrial Edge Hub to the IEM catalog. For more information, see Copying an app from the Industrial Edge Hub to the catalog.
2. Open the IEM catalog.
3. Click "Modbus TCP Connector".
   The latest version app page appears.
4. Click "Update".

5.1.3 Result

After upgrading, open Common Configurator and navigate to "Get Data > Connector Configuration". The latest version of Modbus TCP Connector appears in the list of available connectors.



NOTICE

- After upgrade, tag comments are lost.
- After upgrade, data bus configuration is missing. Manually set the data acquisition cycle, select "Publish on Data Bus", and deploy the configuration to the connector and data bus.
- After upgrade, Modbus TCP Connector version and configuration settings are updated.

Configuring data streams

6

6.1 Managing data sources

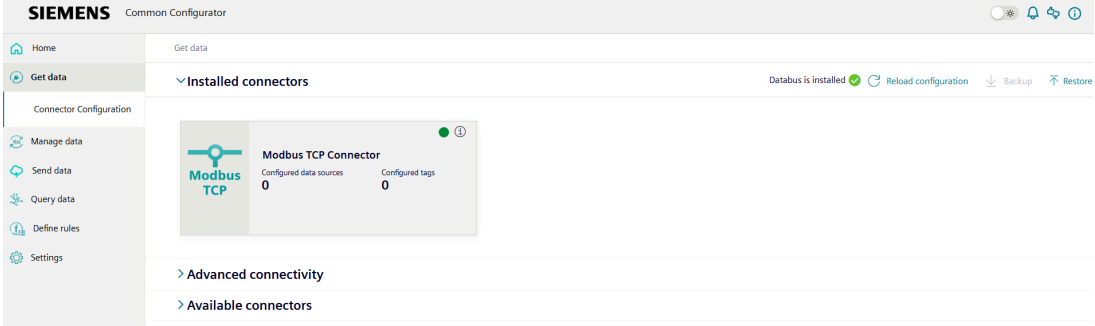
6.1.1 Configuring data sources

A data source is a device that provides data. Use the Common Configurator to configure Modbus TCP Connector to consume data from devices on your IE Device.

6.1.1.1 Accessing the Modbus TCP Connector page

Follow these steps to access the Modbus TCP Connector configuration page:

1. Launch your IE Device and navigate to "Apps > Common Configurator".
2. Click "Get Data > Connector Configuration".
   The "Get Data" page opens and displays all available connectors.



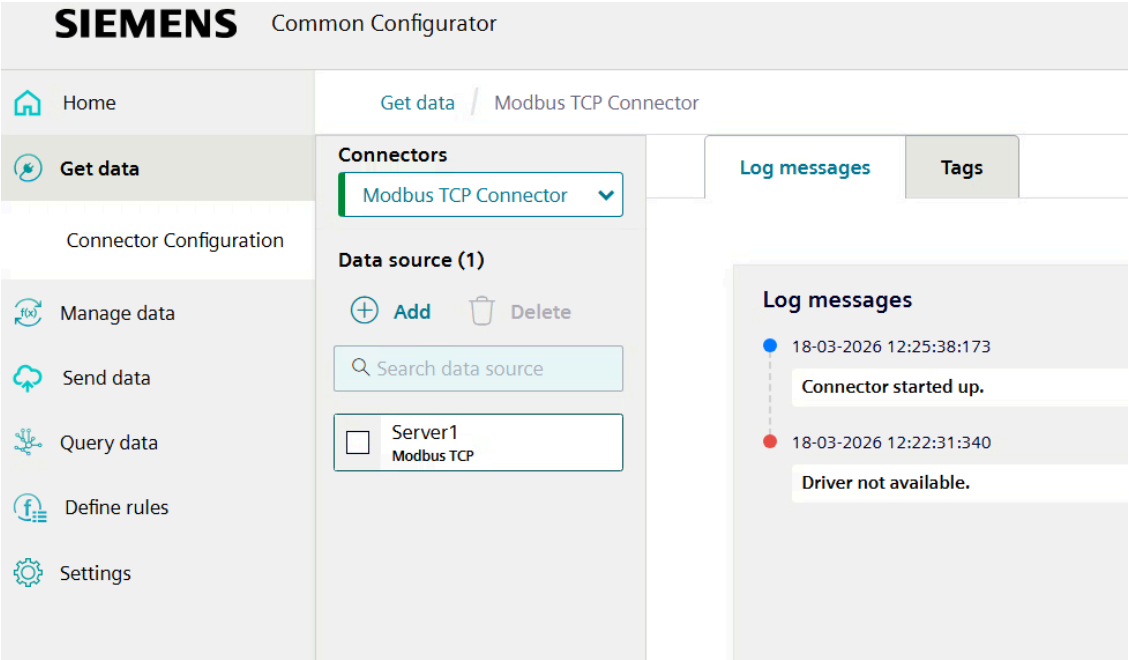
   The page includes the following options:

Option	Description
Reload configuration	Retrieves the connector configuration from the latest deployment and reverts changes made in Common Configurator for all connectors.
Backup	Downloads all connector configurations.
Restore	Imports and restores downloaded configurations.

3. Select "Modbus TCP Connector".
   The Modbus TCP Connector page opens with the following tabs: "Log messages" and "Tags".

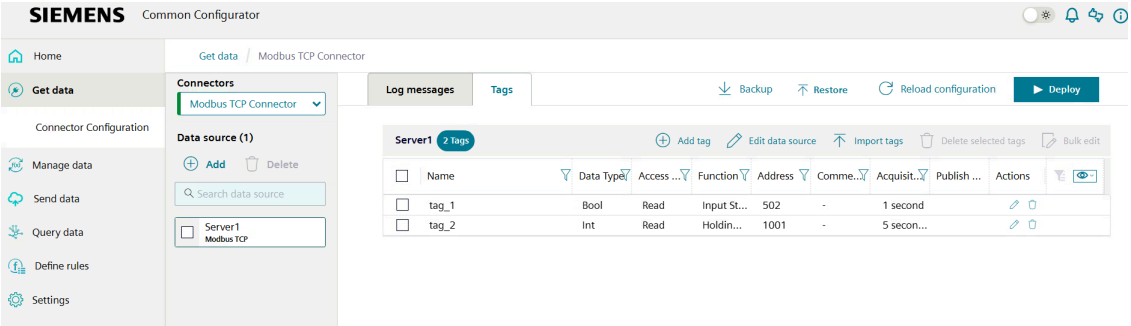
6.1.1.2 Log messages tab

When you open Modbus TCP Connector, the "Log messages" tab displays by default.



6.1.1.3 Tags tab

Click the "Tags" tab to view tag details.



6.1.2 Adding data sources

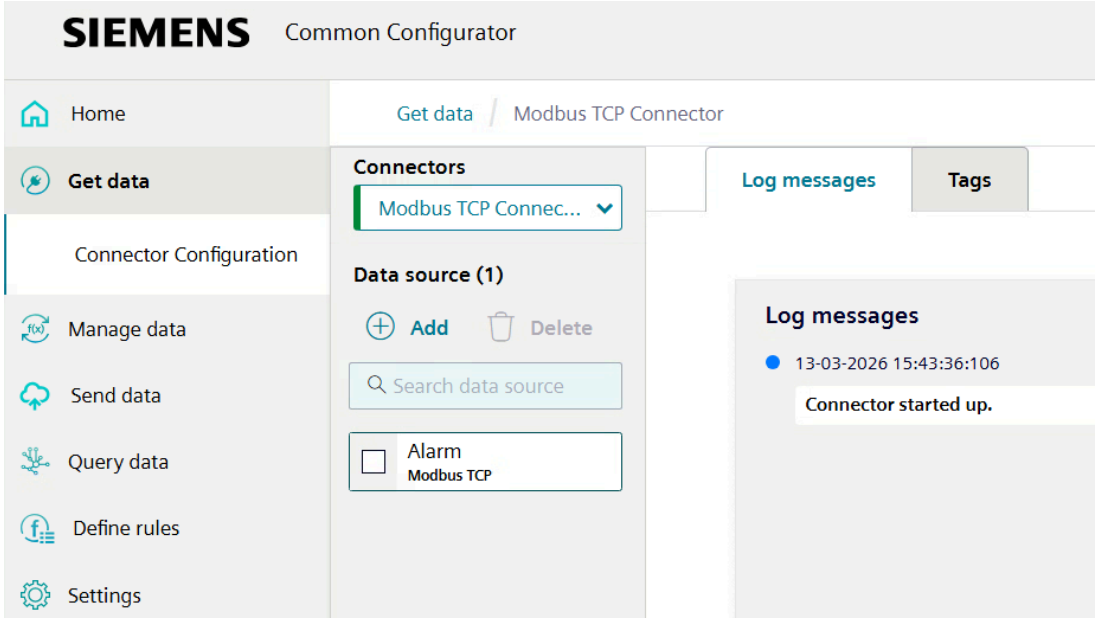
A data source is a device that provides data. You can configure, edit, and delete data sources.

6.1.2.1 Adding a data source

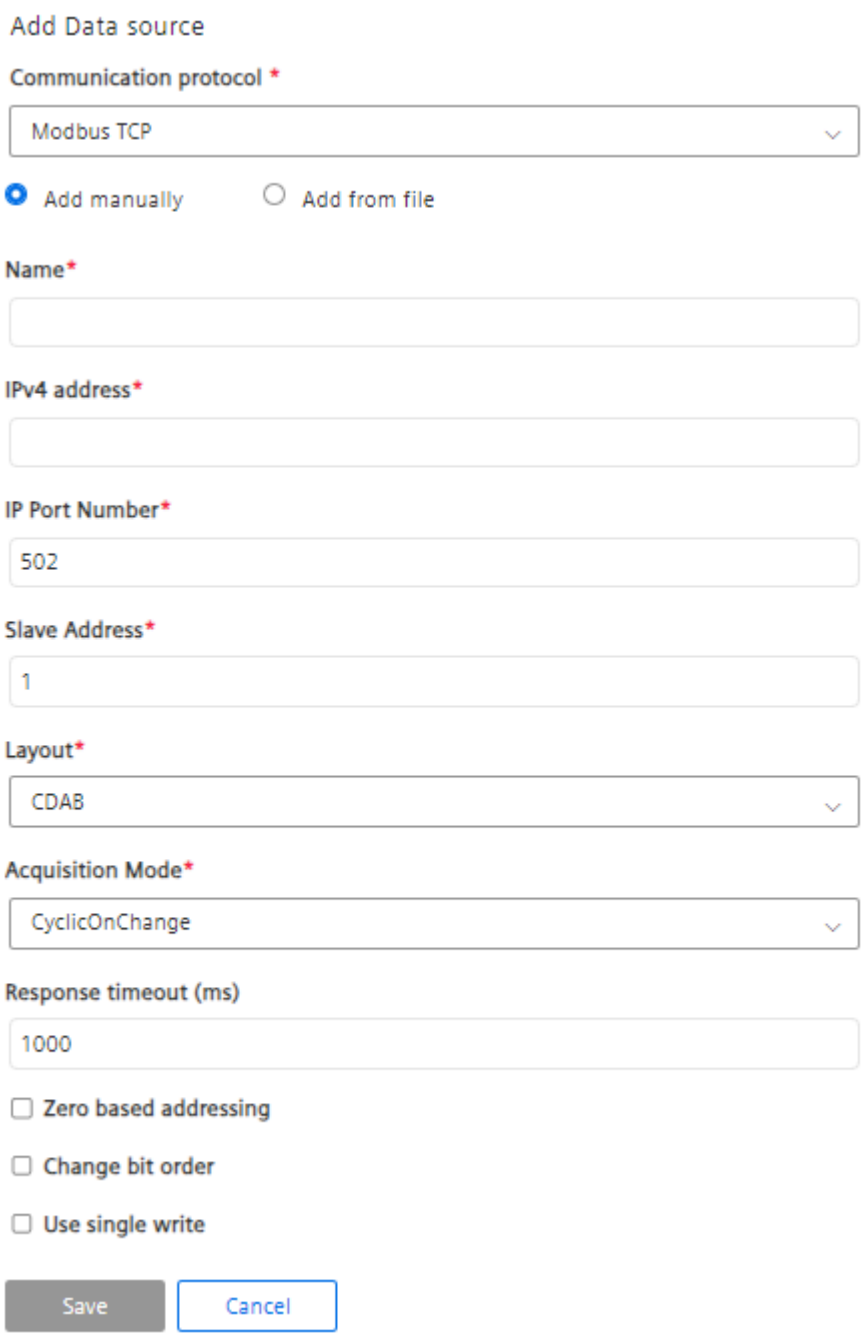
Follow these steps to add a data source:

1. Launch "Common Configurator".
   The "Get data" page opens and displays available connectors.

2. Click "Modbus TCP Connector".
   The "Overview" page of Modbus TCP Connector appears.



3. Click the "+" icon.
   The "Add Data Source" dialog appears.



4. Select "Modbus TCP" from the "Communication protocol" dropdown list.
   Configuration options appear.
5. Enter the following information:

Field	Description
Name	Defines the name of the data source. The name must be unique across all connectors.
IPv4 address	Defines the IP address of the PLC to connect to. Format: decimal, for example, `192.168.1.10`.
IP port number	Defines the port of the IP address for data retrieval. Range: 0–65535, for example, `9600`.
Slave address	Defines the Modbus slave ID.
Layout	Defines the storage format of data in the device. Select big-endian or little-endian for bits and words. Options: ABCD (12345678), BADC (34127856), CDAB (56781234), DCBA (78563412).
Acquisition mode	Select how data is acquired: `CyclicOnChange` updates data when values change; `CyclicOnContinuous` updates data at the defined interval. If the connection interrupts, data continues updating at the interval with Quality Code `0`.
Response timeout in milliseconds	Define the response timeout in ms. Minimum value: `100` ms.
Zero based addressing	Disabled by default. When enabled, configured addresses begin at zero so PLC register and Configurator indices align at zero. When disabled, PLC registers begin at zero and Configurator indices begin at one. The last five digits represent the register offset. Example: address `4x401665` reads Register 10664.
Change bit order	Disabled by default. Affects bit counting in the "Tags" editor. Only applies with "Holding register" function, Boolean data type, and Word data type. For more information, see Changing byte order.
Use single write	Disabled by default. When enabled, function codes 05H, 06H, 15H, and 16H write to the PLC. When disabled, function codes 15H and 16H write to the PLC.

Asterisk (*) indicates mandatory fields.

NOTICE

Add a data source manually or import from a file. Click "Add from File" to import from an exported file. For more information, see Importing configurations

6. Click "Save".
   The data source is added and displayed in the "Data Source" table.

This video can not be displayed in PDF

6.1.3 Editing data sources

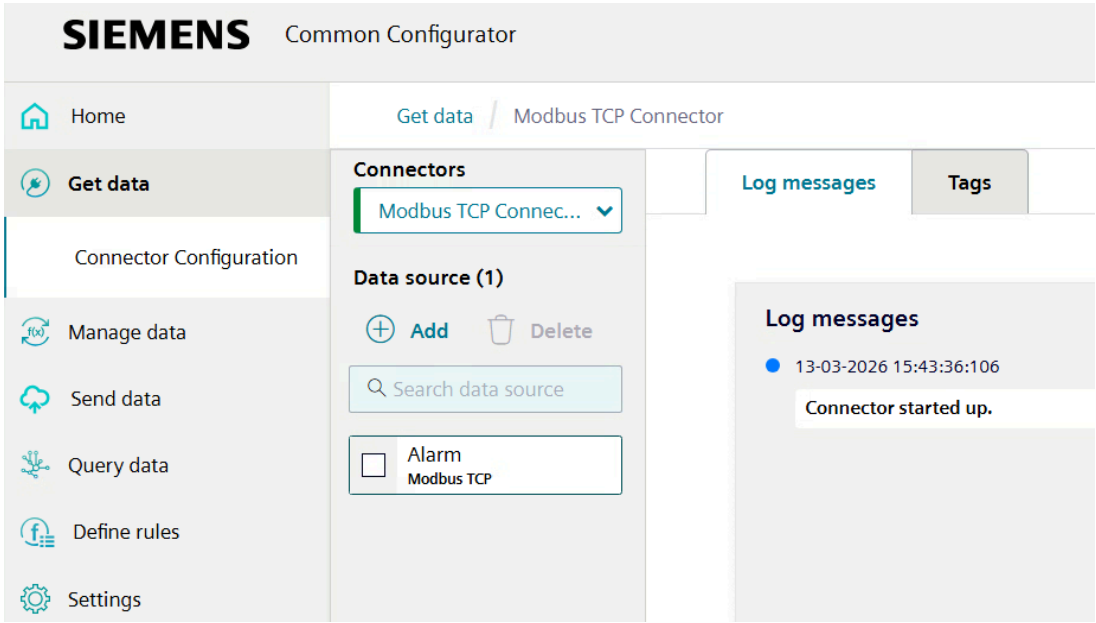
Follow these steps to edit a data source.

1. Open the "Modbus TCP Connector" app in "Common Configurator".
   The "Overview" page appears.
2. Click the "Tags" tab.
   The tag page appears.
3. Click the Edit data source icon to the right of the data source.
   The "Edit Data source" dialog appears.
4. Modify the required fields.
5. Click "Save".
   The data source is updated and displayed in the "Data source" table.

6.1.4 Deleting data sources

Follow these steps to delete a data source.

1. Open the "Modbus TCP Connector" app in "Common Configurator".
   The "Overview" page appears.



2. Select the data source you want to delete.
3. Click the Delete icon.
   A confirmation message appears.
4. Click "Confirm".
   The data source is deleted.

6.2 Managing tags

6.2.1 Adding tags

Tags represent data points from a data source. Use them to read and write data from and to the device.

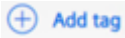
6.2.1.1 Prerequisites

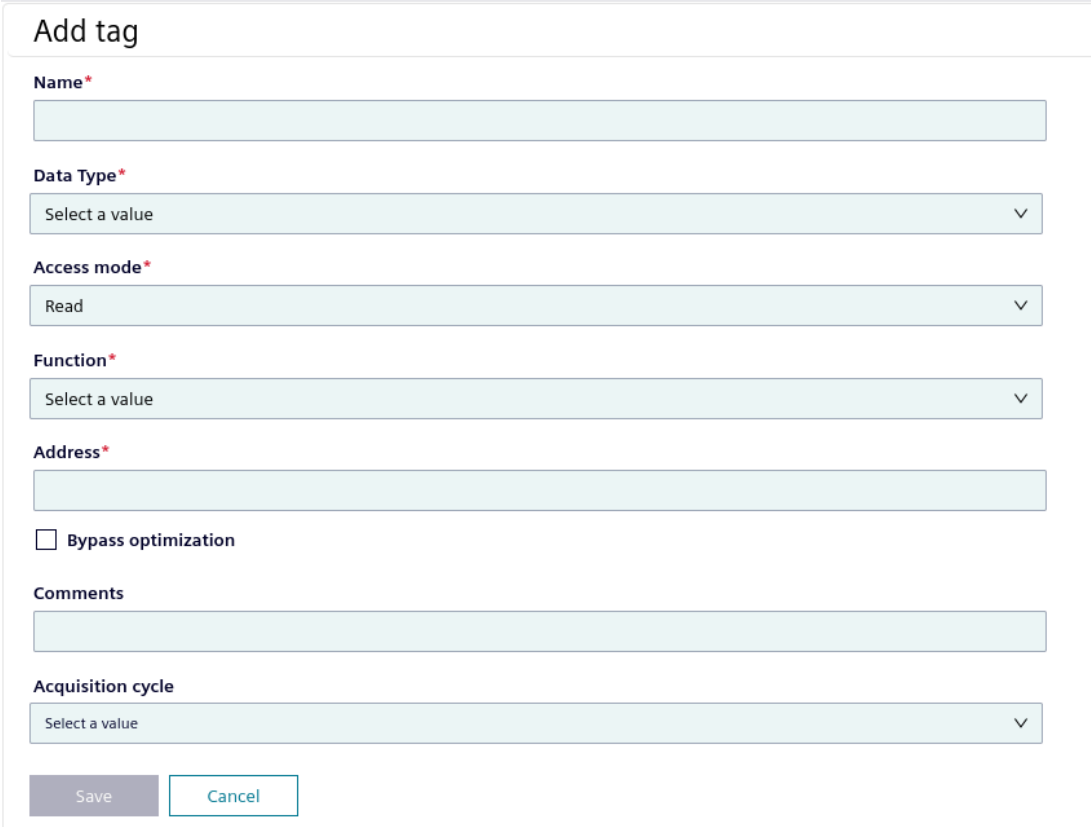
Ensure the following requirements are met before adding tags:

- The Modbus TCP Connector is running.
- A data source is available.

6.2.1.2 Procedure

Follow these steps to add a tag to a data source.

1. Open the "Modbus TCP Connector" app in "Common Configurator".
   The Connector page appears.
2. Click the "Tags" tab.
   The tag page appears.
3. Click the add tag icon to the right of the data source.
   

   The "Add tag" page appears.
   

4. Enter the following information:

Field name	Description
Name	Defines the name of the tag. The name must be unique in the data source.
Data type	Defines the data type of the tag. For more information, see Supported data types.
Access mode	Defines the access mode of the data. The following options are available: Read; Write; Read and write.
Function	The following functions are available: Coil status (0x) — Bool or Word data types, read and write; Input status (1x) — Bool only, read; Input register (3x) — read; Holding register (4x) — read and write; File Record — read.
Address	Defines the tag address. Optionally add H or L to change the byte order. This applies to Word, Int16, UInt16, and String only. The default byte order is L. For register access, enter a direct numeric value. For File Record access, use the notation File Number.Record Number. For more information, see Change byte order.
Bypass optimization	Defines whether to bypass optimization for the tag. If enabled, the tag is read directly instead of being grouped for data acquisition. This option may impact performance. For more information, see Bypass optimization.
Comments	Defines comments for the tag.
Acquisition cycle	Defines the acquisition cycle for sending data. The following options are available: 50 milliseconds; 100 milliseconds; 250 milliseconds; 500 milliseconds; 1 second; 2 seconds; 5 seconds; 10 seconds. The 1 second acquisition cycle is generally used.

All fields are mandatory.

NOTICE

- Support for 50 milliseconds depends on the target hardware (PLC). If the target hardware does not return data in time, the data might be delayed.
- The list of acquisition cycles might vary based on the version of Common Configurator.

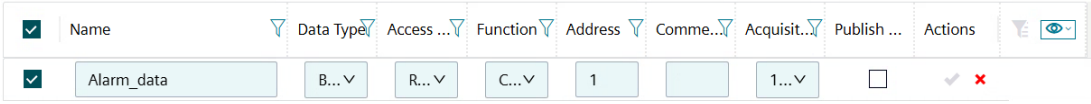
5. Click "Save".
   The tag is added to the data source.

6.2.2 Editing tags

Follow these steps to edit tags:

1. Open the "Modbus TCP Connector" app in "Common Configurator".
   The "Overview" page appears.
2. Click the "Tags" tab.
   The tag page appears.

3. Click the icon on the right side of the tag you want to edit.
   The tag details appear with editable text fields.



4. Modify the required fields.
5. Click the icon.
   The tag is updated and displayed in the "Data source" table.

6.2.3 Deleting tags

Follow these steps to delete a tag.

1. Open the "Modbus TCP Connector" app in "Common Configurator".
   The "Overview" page appears.
2. Click the "Tags" tab.
   The tag page appears.
3. Click the icon on the right side of the tag you want to delete.
   A confirmation message appears.
4. Click "Delete".
   The tag is deleted from the data source.

6.2.3.1 Alternate way to delete tags

1. In the "Tags" tab, select the tags you want to delete.
2. Click the Delete selected tags icon in the upper right corner.
   The selected tags are deleted.

6.2.4 Deploying tags

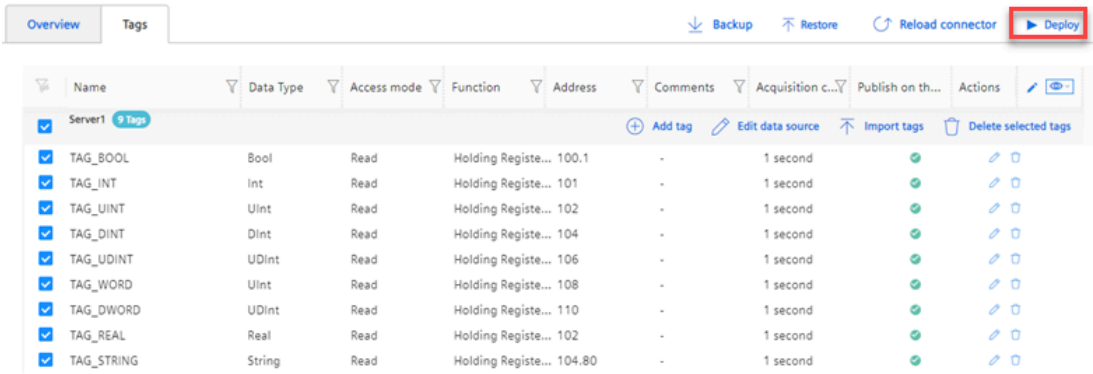
Whenever you make a change in the configuration, deploy the changes so they take effect in the runtime.

6.2.4.1 Procedure

Follow these steps to deploy data points.

1. Select the data sources and data points you want to deploy.

2. Click "Deploy".
   The configuration is saved and loaded to the connector runtime.



6.2.4.2 Result

Once deployed, Common Configurator generates the configuration file for Modbus TCP Connector in the background. This file defines the available data points. The Connector app then publishes data according to the configured acquisition cycle.

6.2.5 Importing configurations

Use the "Import tags" option to import data points and use them in your application.

6.2.5.1 Prerequisites

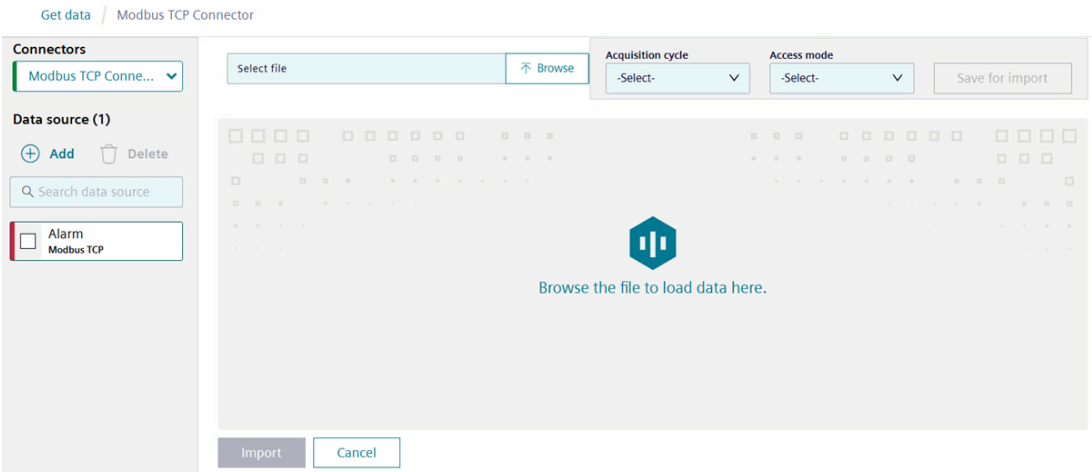
Common Import Converter must be installed on the Industrial IE Device, version 2.1.0-2 or later.

6.2.5.2 Procedure

Follow these steps to import tags at the connection level.

1. Open the "Get Data" tab in Common Configurator.
   Installed and available connectors appear.
2. Click "Modbus TCP Connector".
   The "Overview" page displays whether communication with the PLC has been established.
3. Click the "Tags" tab.
   The tag details appear.

4. Click "Import tags".
   The "Import tag" window appears.



5. Click the browse button and select the file to import.

> **NOTICE**
>
> - You can upload files in either TXT or XLSX format.
> - For XLSX files: After exporting from WinCC TagConverter or TIA Portal, open the file in MS Excel and save it again. The content and type remain unchanged.

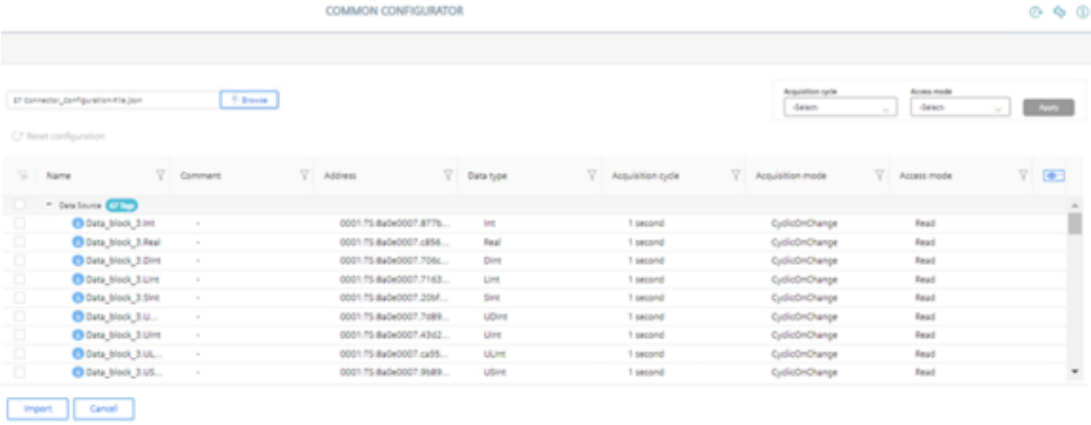
Example TXT file format:

```
TAG_BOOL   %MW100.0   BOOL
TAG_INT    %MW101   INT
TAG_UINT   %MW102   UINT
TAG_DINT   %MW104   DINT
TAG_UDINT  %MW106   UDINT
TAG_WORD   %MW108   WORD
TAG_DWORD  %MW110   DWORD
TAG_REAL   %MW102   REAL
TAG_STRING   %MW104   STRING
```

Example XLSX file format:

Name	Connection	DataType	Address	Start value	Comment	Acquisition mode	Acquisition cycle
TAG_BOOL	PLC	Bit	%MW100.0			Visible	1 s
TAG_INT	PLC	± Int	%MW101			Visible	1 s
TAG_UINT	PLC	Int	%MW102			Visible	1 s
TAG_DINT	PLC	± Double	%MW104			Visible	1 s
TAG_UDINT	PLC	Double	%MW106			Visible	1 s
TAG_WORD	PLC	Int	%MW108			Visible	1 s
TAG_DWORD	PLC	Double	%MW110			Visible	1 s
TAG_REAL	PLC	Float	%MW102			Visible	1 s
TAG_STRING	PLC	ASCII	%MW104			Visible	1 s

6. Select the acquisition cycle and click "Apply".
   The tags included in the exported project appear.



NOTICE

- Tags that cannot be imported appear under "Invalid Tags".
- If tags are imported from an unknown data source, a new data source is created.
- If a data source with the same name already exists, the imported tags are added to existing tags.
- If the data source with tags is already configured, all tags appear under the configured data source section.
- For text files: Pay attention to the format of each row and how each field is separated. If the format does not match, the import fails.

7. Select the tags you want to publish.
8. Click "Import".
   Repeat the procedure for additional tags with a different acquisition cycle.

6.2.5.3 Result

The imported tags appear in the "Tags" page. Click "Deploy" to publish data and metadata. For more information, see Deploying tags.

6.3 Configuring via Databus (Optional)

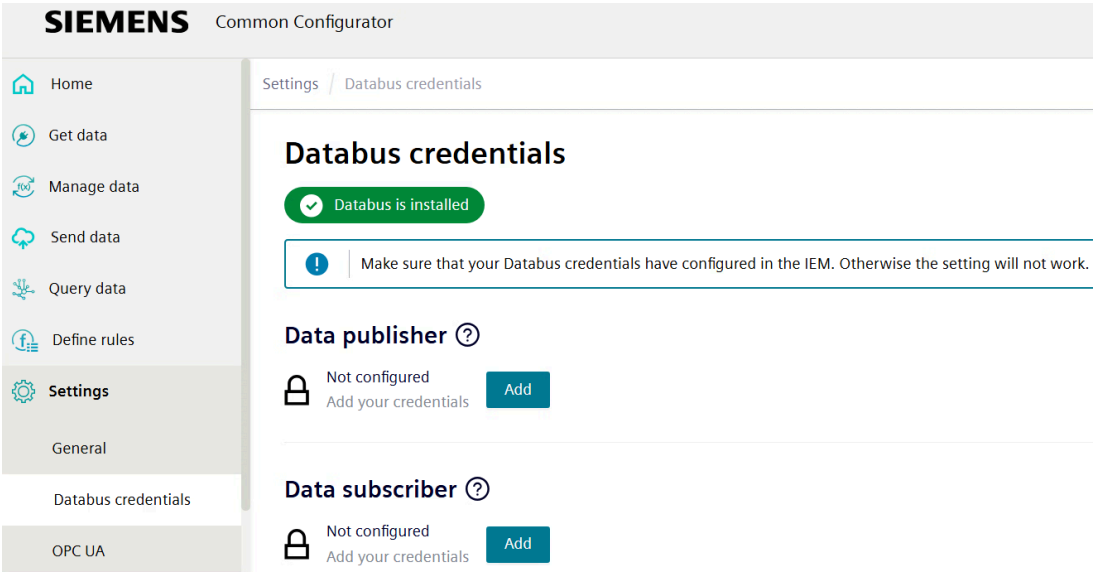
6.3.1 Entering Databus credentials in Common Configurator

To publish Modbus TCP data in the Databus, enter the Databus credentials in the Common Configurator.

6.3.1.1 Entering the credentials

Follow these steps:

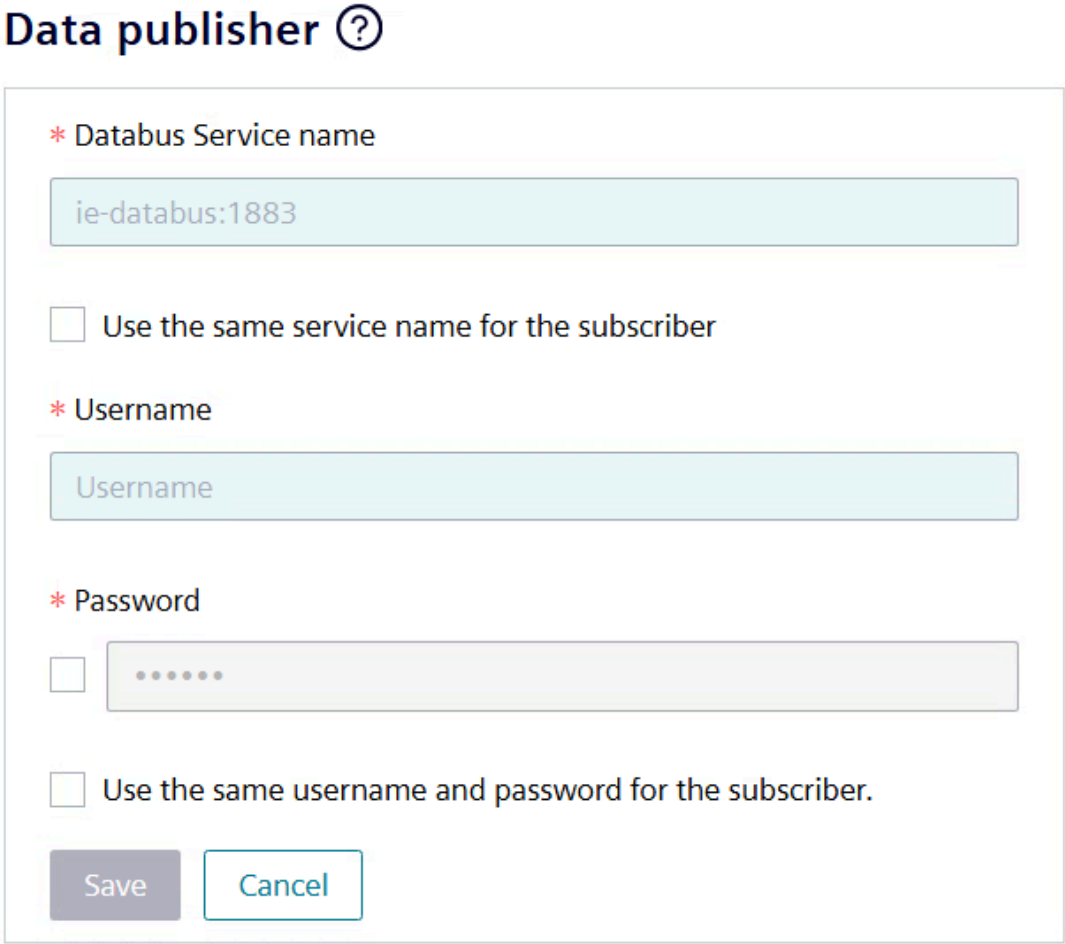
1. Open "Settings" > "Databus Credentials" tab.
   The "Databus credentials" page appears.



NOTICE

Make sure that your Databus credentials have configured in the IEM. Otherwise the setting will not wwork.

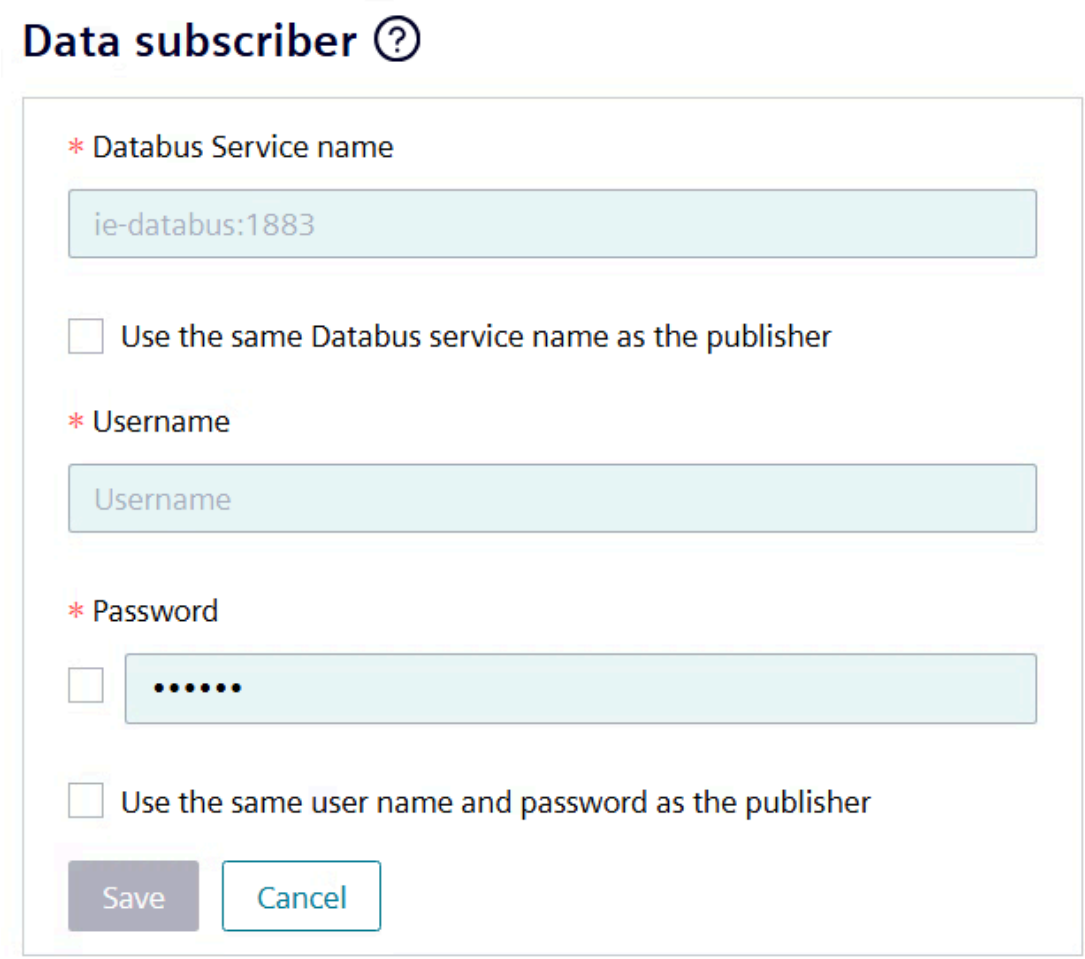
2. Click "Add" on the "Data Publisher" tab and enter the corresponding credentials.



3. Enter the Databus service name, e.g. `ie-databus:1883`.

4. Click "Save".

5. Click "Add" on the "Data Subscriber" tab and enter the corresponding credentials.



6. Click "Save".

6.3.2 Configuring metadata topics and users in Databus

The Databus system app exchanges data between a PLC and the IE Device. The Modbus TCP Connector sends data to the Databus on the IE Device for further processing.

You must create a user and one or more topics containing the Modbus TCP data in the Databus configuration. The following topics are required:

- `ie/m/j/simatic/v1/mbtcp1/dp` for Modbus TCP metadata
- `ie/d/j/simatic/v1/mbtcp1/dp/#` for Modbus TCP data

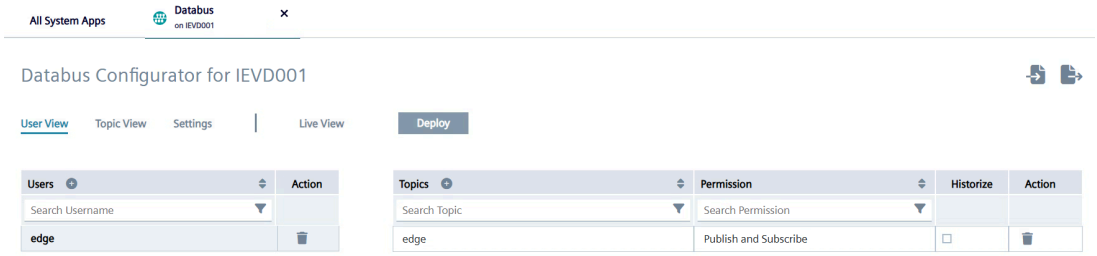
Before you start, ensure you are logged into the IEM.

6.3.2.1 Configuring Databus

To configure Databus, follow these steps:

1. Open the "Data Connections" tab.
2. Click the "Databus" tile.
3. Select the appropriate IE Device.

4. Click the plus icon (+) next to "Users" to create a user.
5. Define the username and password.
6. Click the plus icon (+) next to "Topics" to create a topic.
7. Enter `ie/#` as the topic name.
8. Under "Permission", define the "Publish and Subscribe" permission.
   The configuration looks like this:



9. Click "Deploy" to deploy the configuration.

> **NOTICE**
>
> More information on the configuration of the Databus can be found here: Databus configuration

6.3.2.2 Result

You have configured Databus and deployed the configuration.

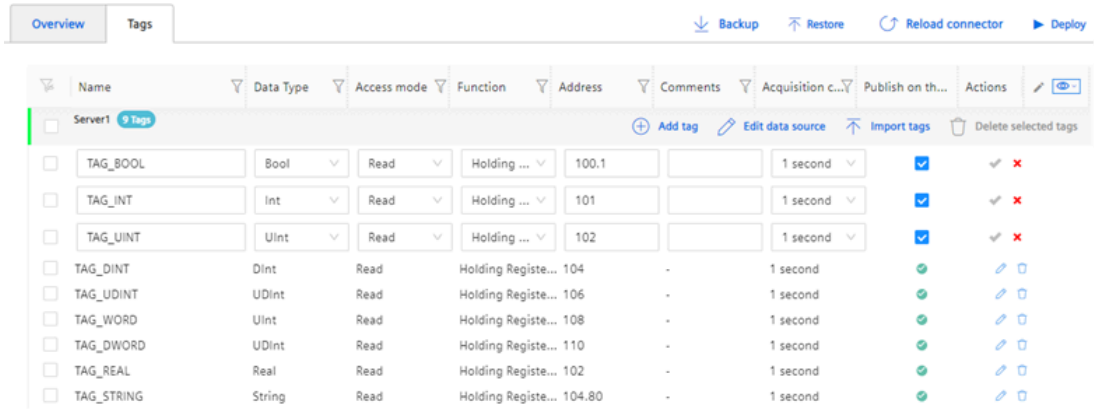
6.3.3 Deploying Modbus TCP tags in Databus (Common Configurator)

Modbus TCP tags become available in the Databus after you select the corresponding option for the selected tags.

6.3.3.1 Deploying tags to Databus

To deploy tags to the Databus, follow these steps:

1. Select the "Publish to Databus" option for all Modbus TCP tags or for individual selected tags.



2. Click "Deploy".
   The configuration file for the Databus is created automatically.

6.3.3.2 Result

In the background, Industrial Information Hub (IIH) generates the configuration file for the Modbus TCP Connector and sends it to the connector. This file defines which tags are available for the Databus.

The configuration files are located in the "Management" tab of the IE Device.

> **Credentials for Databus**
>
> Enter the Databus credentials in the settings to enable the connection. For more information, see Enter credentials for Databus (Common Configurator).

Limitations 7

7.1 Limitations of Modbus TCP Connector

7.1.1 Databus Gateway

- Only JSON format `timeseries` is supported. The `bulk` and `binary` formats are not supported.
- A reboot is required after deploying a new configuration. Communication is interrupted during this time.

7.1.2 Modbus TCP Connector

- A reboot is required after deploying a new configuration. Communication is interrupted during this time.
- Structured data types and PLC data types (UDT) are not supported. Select each individual element of a structure or PLC data type.
- Individual read tags include the IE Device timestamp.
- Modbus TCP Connector supports reading only.
- When the Bool data type is used with bit registers (the "Coil status" and "Input status" functions), the address is the corresponding register address. When the Bool data type is used with the "Holding register" function, the address is the corresponding register address plus bit offset; for example, `100.5`.
- The Word data type works only with the "Coil status" and "Holding register" functions.
- Cyclic continuous data acquisition sends all data, even if the data does not change. This behavior is required for use cases like machine learning. Enabling this mode impacts IE Device performance:
  - Increased load on northbound GRPC connections: The Databus and IIH Essentials experience higher traffic from the continuous data stream.
  - Higher storage usage: More storage is consumed for time-series data in IIH Essentials.
  - No impact on PLC communication: The PLC connection operates independently on a polling mechanism.

7.1.3 Databus Gateway and Modbus TCP Connector

The following limitations apply when both apps are used together:

Function	Supported by Databus Gateway
Subscribe to data points	Yes
Write data points	Yes
Read data points	No
Driver status events	Yes
PLC alarms	No

7.1.4 Connectivity Suite Connectors

- Diagnostics are limited to the app log file.
- Datasets are not supported.
- Export XLSX files from TIA Portal or WinCC TagConverter, then open and save the file in MS Excel. The file content and type remain unchanged.

7.1.5 Common Configurator

Dedicated device data types are not supported because Databus Gateway uses standard Edge data types. This limitation applies only if you create the subscription configuration using Common Configurator. If you create the configuration manually, no restrictions apply. For more information on standard Edge data types, see Supported data types.

Performance measures

8

8.1 Performance results

8.1.1 Test environment

- Siemens SIMATIC CPU 1518F-4 PN/DP MFP
- IE Device SIMATIC IPC 427E IE Device
- Modbus TCP Connector V3.0.0
- GRPC client/server

8.1.2 Test scenario

Number of tags of data type Int16	40,000
Rate of tag change	The values of all tags change every 2 seconds.
Configured update cycle	1 second
Test client	GRPC client retrieves messages from the GRPC server in the connector.

8.1.3 Test result

The test is successful. 40,000 value changes per second of data type Int16 are successfully transmitted, with no value changes lost.

Appendix 9

9.1 Quality codes

The quality code characterizes the quality of a value when it is transferred from a PLC to Modbus TCP Connector.

Using the PLC time stamp in Modbus TCP Connector affects the quality code.

Flag (bits 12 ... 15)	Name	Description
Bit 12	Source quality	-
Bit 13	Source time	The data time stamp was created and added by an external data source.
Bit 14	Time corrected	The data time stamp added by an external data source is corrected in the system.
Bit 15	Reserved	-

9.2 Change bit order

---	---	---	---	---	---	---	---	---	---	---
	Bit order enabled	Big-endian								
Bit address	16	15	14	13	12	11	10	9	8	7
	0	0	0	0	0	0	0	0	0	1
---	---	---	---	---	---	---	---	---	---	---
	Bit order disabled	Little-endian								
Bit address	16	15	14	13	12	11	10	9	8	7
	0	0	1	0	0	0	1	0	0	0

9.3 Changing byte order

You can optionally add `H` or `L` to a tag's address field to change the byte order.

For example, if the function is "Holding Register" and the address is `100`, adding `H` or `L` results in `100H` or `100L`. By default, address `100` is equivalent to `100L`.

This function applies only to the following data types:

Data type	Effect
Word / Int16 / UInt16	Swaps the byte order. For example, value `0x1234` without the `H` option remains `0x1234`. Adding `H` to the address field swaps the bytes, resulting in `0x3412`.
String	Swaps the byte order within each two-character pair. For example, string `abcdefgh` without the `H` option remains unchanged. Adding `H` results in `badcfehg`.